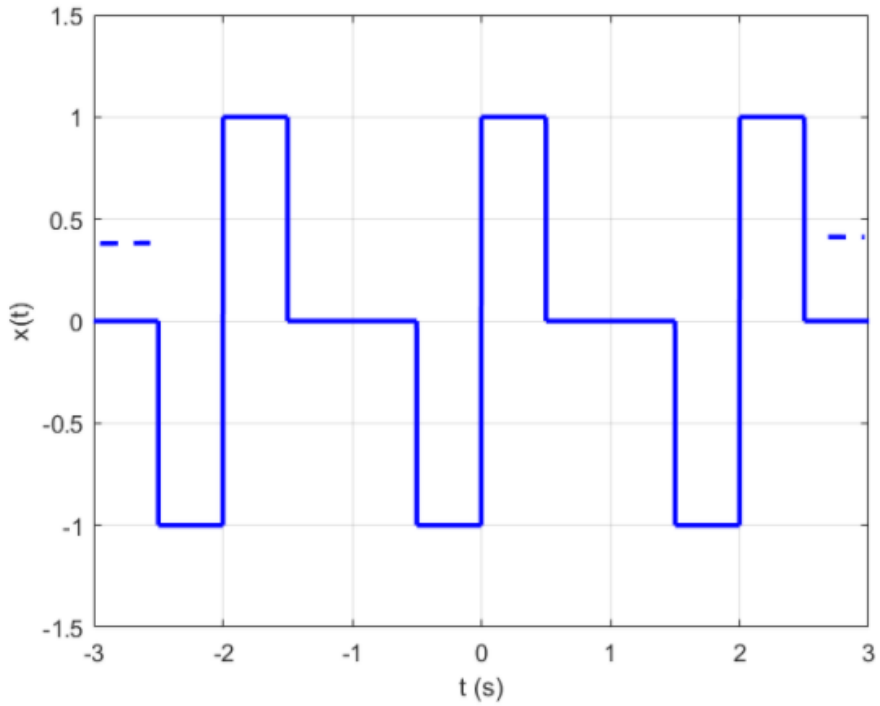
QUESTION 1: MULTIPLE CHOICE

15 out of 15 points

Consider the signal x(t) shown below. Which one of the below expressions corresponds to the FS coefficient X[K]?



Given Answer: $X[k] = \frac{j}{\pi k}\left[\cos\left(\frac{\pi k}{2}\right) - 1\right]$

Correct Answer: $X[k] = \frac{j}{\pi k}\left[\cos\left(\frac{\pi k}{2}\right) - 1\right]$

QUESTION 2: MULTIPLE CHOICE

15 out of 15 points

Consider a periodic signal x(t) with period 0.05 s whose FS coefficients are given by:

$$X[k] = \begin{cases} 0 & k = 0 \\ |k| & k \text{ odd} \\ 3jk & k \neq 0 \text{ even} \end{cases}$$

x(t) is applied to the input of a filter whose transfer function is given by H(f)=3tri(f/60). Which one of the below expressions corresponds to the output signal y(t)?

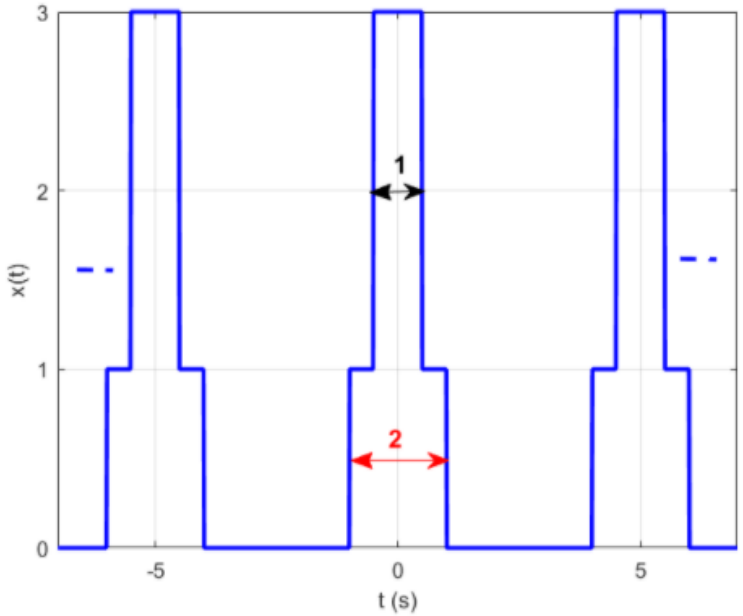
Given Answer: $4\cos(40\pi t) - 12\sin(80\pi t)$

Correct Answer: $4\cos(40\pi t) - 12\sin(80\pi t)$

QUESTION 3: CALCULATED NUMERIC

5 out of 5 points

Consider the signal x(t) shown below.
The FS coefficients of x(t) can be written as X[k]=a×sinc(bk)+c×sinc(dk) where a, b, c and d are positive constants. If b<d, what is the value of d?



Given Answer: 0.4

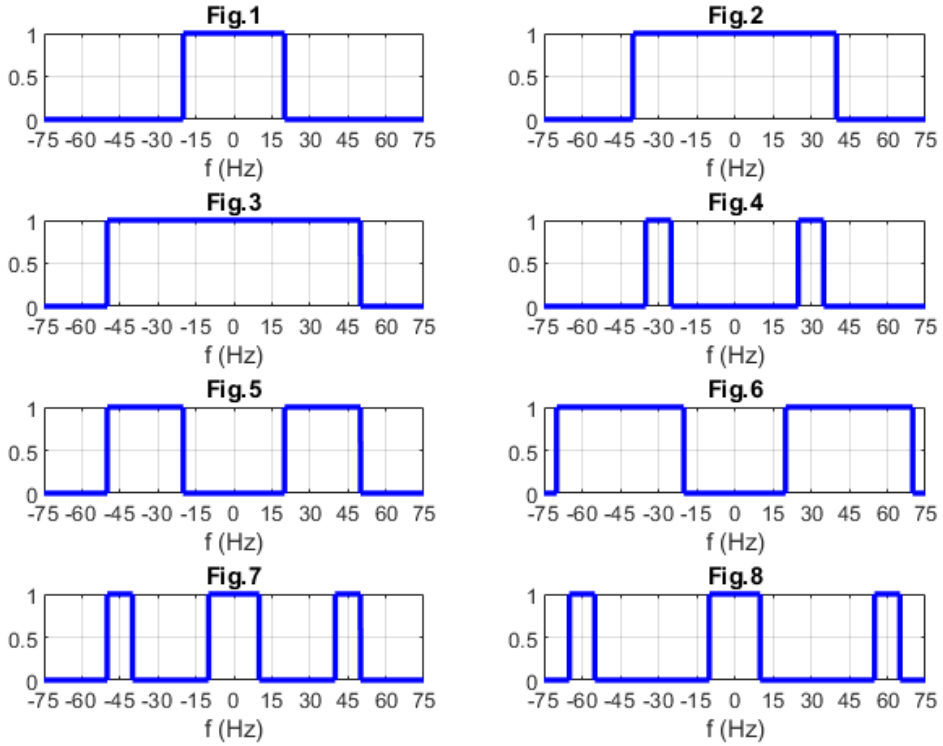
Correct Answer: 0.4

Answer range +/- 0 (0.4 - 0.4)

QUESTION 4: MULTIPLE CHOICE

5 out of 5 points

A periodic signal with fundamental frequency 15 Hz is applied to the input of a filter and the output signal is given by y(t)=a×cos(60πt)+b×cos(90πt)+c×cos(120πt). Which one of the below figures corresponds to the transfer function of the filter?



Given Answer: Fig. 6

Correct Answer: Fig. 6

QUESTION 5: CALCULATED NUMERIC

10 out of 10 points

Consider a periodic signal whose nonzero trigonometric FS coefficients are given by:

- $X_c[0]=2$
- $X_c[1]=3$
- $X_c[3]=1$ and $X_s[3]=2$

What is the power of the signal?

Given Answer: 11

Correct Answer: 11

Answer range +/- 0 (11 - 11)

QUESTION 6: TRUE/FALSE

5 out of 5 points

Consider a signal whose Fourier Series coefficients are given by:

$$X[k] = \begin{cases} jk & -4 \le k \le 4 \\ 0 & \text{otherwise} \end{cases}$$

The signal is real.

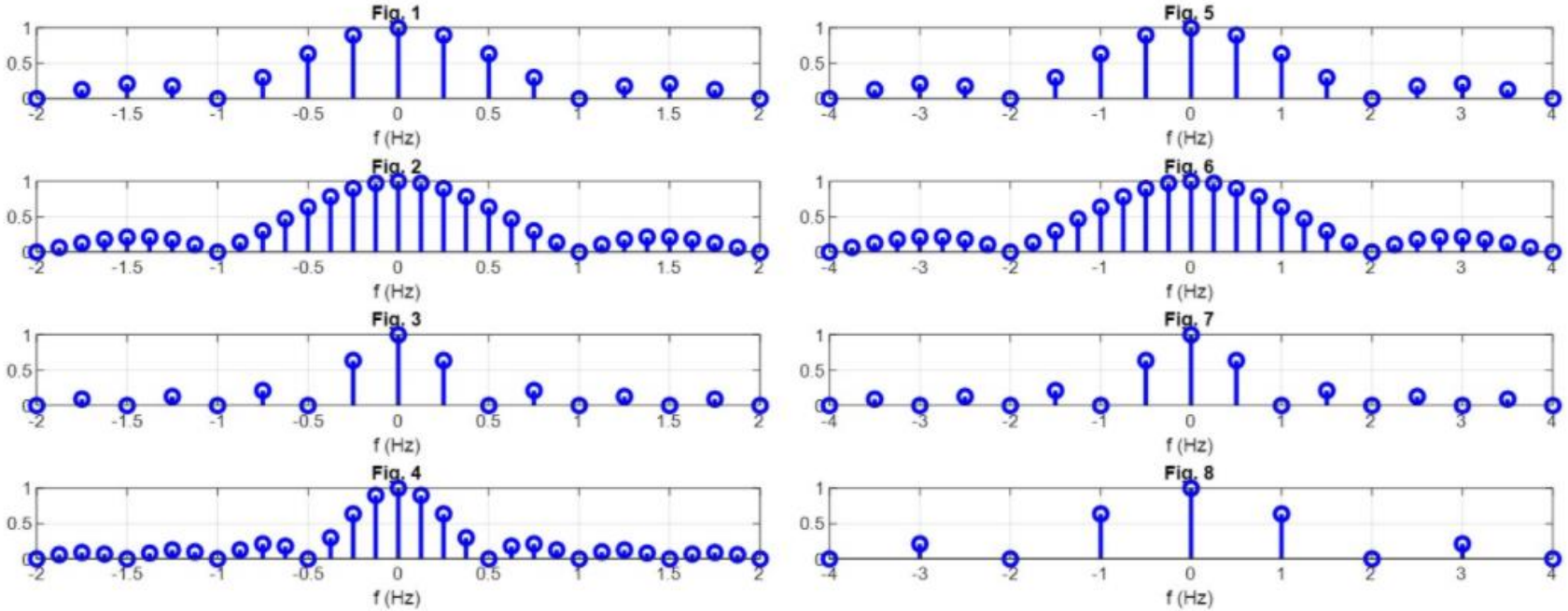
Given Answer: True

Correct Answer: True

QUESTION 7: MULTIPLE CHOICE

10 out of 10 points

Which one of the below figures corresponds to the magnitude spectrum of the signal $x(t) = \frac{1}{2}\mathrm{rect}\left(\frac{t}{2}\right) * \mathrm{comb}\left(\frac{t}{4}\right)$?



Given Answer: Fig. 3

Correct Answer: Fig. 3

QUESTION 8: MULTIPLE CHOICE

5 out of 5 points

Consider the signal given by:

$$x(t) = 3 + \cos(150\,\pi t) + 4\cos(300\,\pi t) + 6\cos(450\,\pi t)$$

What is the value of X[2]?

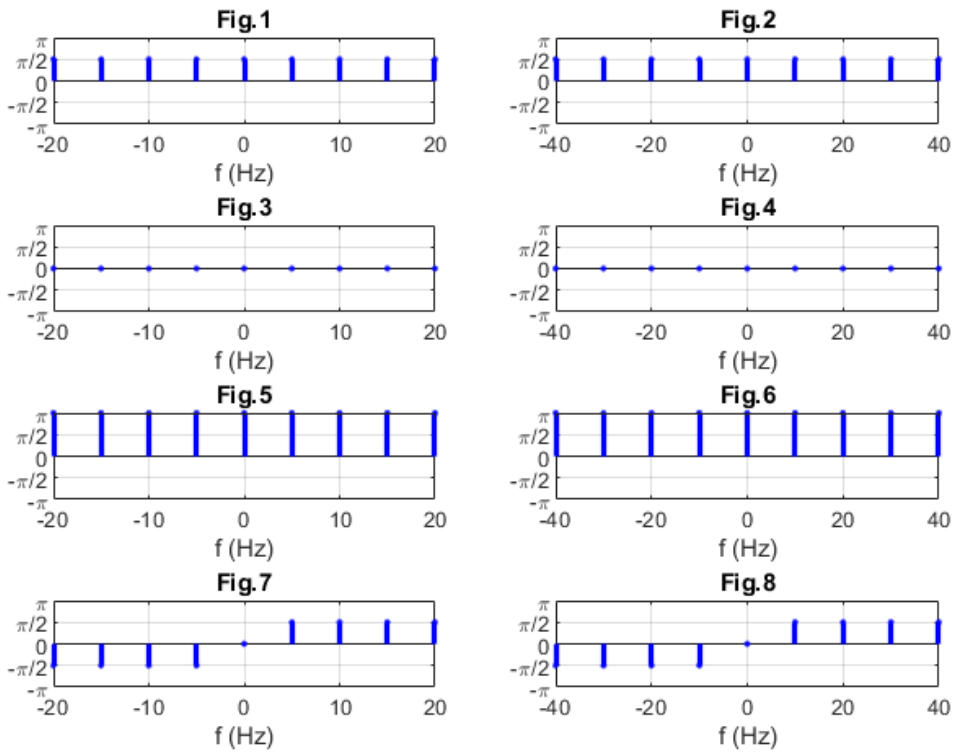
Given Answer: 2

Correct Answer: 2

QUESTION 9: MULTIPLE CHOICE

5 out of 5 points

Consider a real and odd periodic signal with period 0.2 sec. Which one of the below figures corresponds to the phase spectrum of the signal?



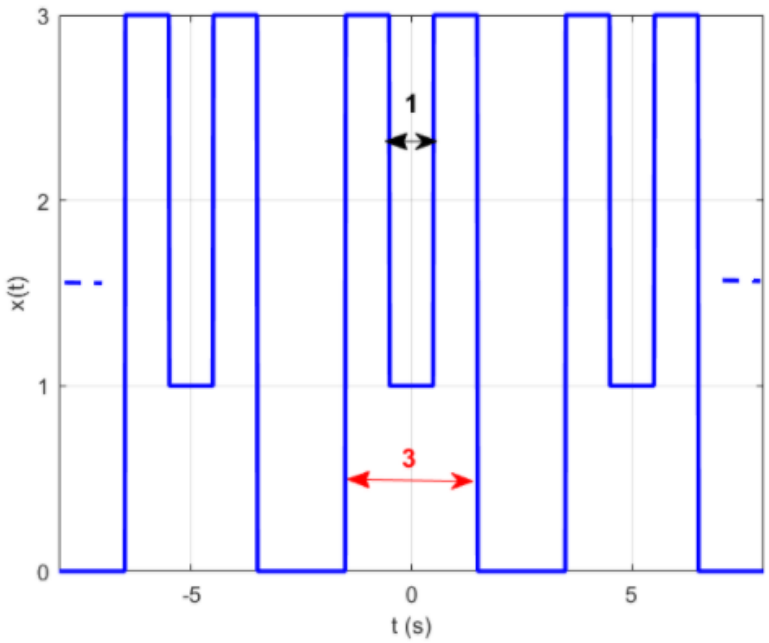
Given Answer: Fig. 7

Correct Answer: Fig. 7

QUESTION 10: CALCULATED NUMERIC

5 out of 5 points

Consider the signal x(t) shown below.
The FS coefficients of x(t) can be written as X[k]=a×sinc(bk)-c×sinc(dk) where a, b, c and d are positive constants. What is the value of a?



Given Answer: 1.8

Correct Answer: 1.8

Answer range +/- 0 (1.8 - 1.8)

QUESTION 11: MULTIPLE CHOICE

5 out of 5 points

Consider the signal given by:

$$x(t) = \frac{1}{2}\sin(200\,\pi t) + \sin(600\,\pi t)$$

What is the power of the first harmonic?

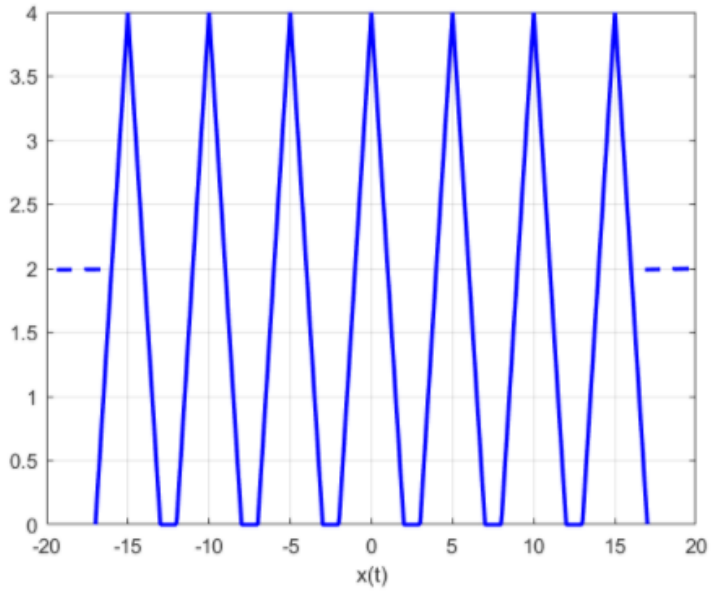
Given Answer: 1/8

Correct Answer: 1/8

QUESTION 12: CALCULATED NUMERIC

5 out of 5 points

Consider the periodic signal shown below. If the signal can be written as x(t)=a×tri(t/b)*comb(t/c), what is the value of the constant a?



Given Answer: 0.8

Correct Answer: 0.8

Answer range +/- 0 (0.8 - 0.8)

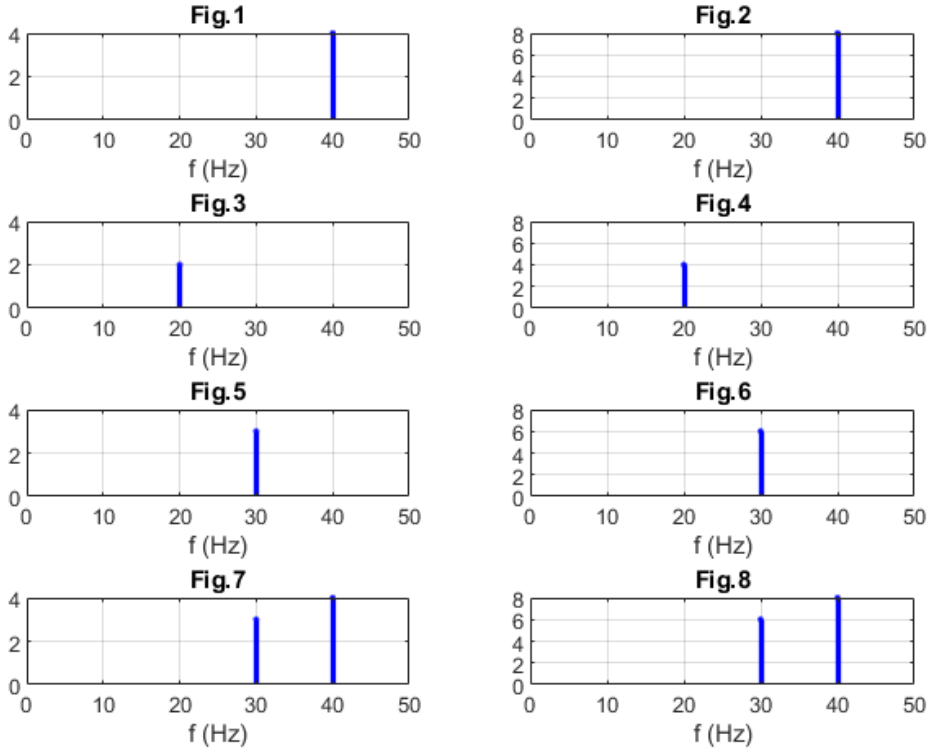
QUESTION 13: MULTIPLE CHOICE

10 out of 10 points

Consider a periodic signal with period 0.1 sec whose Fourier Series coefficients are given by:

$$X[k] = \begin{cases} |k|/2 & -5 \le k \le 5 \\ 0 & \text{otherwise} \end{cases}$$

The signal is applied to the input of an ideal bandpass filter with a lower cutoff frequency of 25 Hz and upper cutoff frequency of 35 Hz. If the filter introduces a gain of 2, which one of the below figures corresponds to the magnitude spectrum of the output signal (for positive frequencies)?



Given Answer: Fig. 5

Correct Answer: Fig. 5